97.1
Pass

Ravi Kumar

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Candidate Information

Email	ravikumar80nitj@gmail.com
Test	eClerx - Java Coding Test - 3+ Years New
Candidate Packet	View
Taken on	24 Feb 2026 22:43:31 IST
Time taken	88 min 18 sec/ 90 min
Work Experience	3 years
Invited by	Vipul
Invited on	24 Feb 2026 18:11:53 IST

Skill Distribution

No.	Skill	Score
1	Spring Boot Advanced	100%

2	Spring Boot Intermediate	75%
3	Problem Solving Intermediate	100%
4	Java Intermediate	100%

Tags Distribution

Hard	100%	Spring Boot	80%
TDD	100%	Spring	50%
JPA	0%	Spring Data	0%
Transactional	0%	Medium	97%
Application Properties Annotations	100%	Exception Handling	100%
Aspect-Oriented Programming	100%	Implementation	100%
Arrays	100%	Algorithms	100%
Problem Solving	100%	Greedy	100%
Sorting	100%	Java	100%

Questions

Spring Boot (Advanced) • 20 / 25

No.	Question	Skill	Score	Code Quality	Time Taken	Status

1	Output Multiple Choice	Spring Boot (Advanced)	5/5	-	7 min 39 sec	✓
2	Spring Data Transactional Multiple Choice	Spring Boot (Intermediate)	0/5	-	1 min 32 sec	✗
3	Configuration Properties Multiple Choice	Spring Boot (Intermediate)	5/5	-	3 min 46 sec	✓
4	Endpoint Security Multiple Choice	Spring Boot (Intermediate)	5/5	-	1 min 17 sec	✓
5	Predict the Output Multiple Choice	Spring Boot (Intermediate)	5/5	-	4 min 39 sec	✓

Problem Solving (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status
6	Wind Farms Coding	Problem Solving (Intermediate)	75/75	-	53 min 46 sec	✓

Java (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status

7	Java: DataVille's Optimal Storage Coding	Java (Intermediate)	75/75	-	14 min 13 sec	✓

1. Output

Correct

Multiple Choice | Hard | Spring Boot | TDD

Question description

What will happen to the output file?

```
@SpringBootTest
public class LogTest
{
   File log;
   FileWriter logWriter;

   @Autowired
   SomeClassToTest someClassToTest;

   @Before
   public void init()
   {
     log = new File("log.txt");
     logWriter = new FileWriter(log);
   }

   @After
   public void teardown()
   {
      log.delete();
   }

   @Test
   public void test1()
   {
      logWriter.write("Entered Test 1");
      assertTrue(someClassToTest.someMethod());
   }
```

```
    @Test
    public void test2()
    {
        logWriter.write("Entered Test 2");
        assertTrue(someClassToTest.otherMethod());
    }
}
```

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ The output file is not created.
- ○ The output file is created but not deleted.
- ● **The output file is created and eventually deleted.** ✓
- ○ The create and delete blocks will not be executed.

No comments.

2. Spring Data Transactional

Incorrect

Multiple Choice | Spring | Spring Boot | JPA | Spring Data | Transactional | Medium

Question description

Given the following code, what happens when *StudentRepsitoryTest* runs?

Lesson.java

```
@Entity
public class Lesson {
   @Id
   private Long id;
   private String name;
   private Integer courseLength;
   private Integer courseLevel;
   @JsonBackReference
   @ManyToOne(fetch = FetchType.LAZY)
   @JoinColumn(name = "student_id")
   private Student student;
   private boolean inProgress;
   //getter & setters
}
```

Student.java

```
@Entity
public class Student {
   @Id
   private Long id;
   private String name;
   private String surname;
   @JsonManagedReference
   @OneToMany(mappedBy = "student", fetch = FetchType.EAGER, cascade = {CascadeType.PERSIST,
CascadeType.REFRESH})
   private List<Lesson> lessons;
   //getter & setters
}
```

StudentRepository.java

```
@Repository
public interface StudentRepository extends JpaRepository<Student, Long>{
   //Y
```

```
    Optional<Student> findById(Long id);
}
```

LessonRepository.java

```
@Repository
public interface LessonRepository extends JpaRepository<Lesson, Long> {
}
```

StudentService.java

```
@Service
public class StudentService{
    private StudentRepository studentRepository;
    public StudentService(StudentRepository studentRepository, LessonRepository lessonRepository)
{
        this.studentRepository = studentRepository;
        this.lessonRepository = lessonRepository;
    }

    public void save(Student student) {
        studentRepository.save(student);
    }

    //other crud metods

    //X
    public void deleteStudentWithLessons(Long id) throws SQLException {
        Student student =
                studentRepository.findById(id).orElseThrow(() -> new RuntimeException("Student not foun
d"));
        List<Lesson> lessons = student.getLessons();
        lessons.stream().forEach(lesson -> lesson.setStudent(null));
        studentRepository.delete(student);

        boolean inProgress =lessons.stream()
                .map(Lesson::isInProgress)
                .filter(progress -> progress == true)
                .findFirst()
                .orElseGet(() -> false);
        if (inProgress) {
```

```
            throw new SQLException("Cannot delete all Lessons. There are lessons in progress state");
        }
        lessonRepository.deleteAll(lessons);
    }
}
```

StudentRepositoryTest.java

```
@SpringBootTest
class StudentRepositoryTest {
    @Autowired
    StudentRepository studentRepository;
    @Autowired
    StudentService studentService;
    @Autowired
    LessonRepository lessonRepository;
    Student student;
    @BeforeEach
    public void init() {
        student = Student.builder().name("Jorn").surname("Deynhoven").build();
        studentRepository.save(student);
        lessonRepository.saveAll(List.of(
                Lesson.builder().name("Music").courseLength(10).student(student).courseLevel(123).build
(),
                Lesson.builder().name("Art").courseLength(20).courseLevel(123).student(student).build(),
                Lesson.builder().name("Trance").courseLength(30).courseLevel(123).student(student).inPro
gress(true).build()
        ));
    }
    @Test
    void deleteStudentWithLessonsTest() throws SQLException {
        studentService.deleteStudentWithLessons(student.getId());
    }
}
```

Interviewer guidelines

Candidate has to know how to use spring data in any spring application which uses spring data. If an entity wanted to remove from database without removing the child attributes, then first relation between them should be removed. Without doing this, spring will throw DataIntegrityViolationException to keep database consistent. To achive that without changing existing code, we need to

annotate deleteStudentWithLessons(Long id) method with @Transactional annotation. With this spring will update the database states of child entities. In this case it will update Lesson attributes to have student as null. Then student will be removed in the next line.

After this there is a inProgress check for existing lessons. it will throw SQLException which is a checked exception. @Transactional annotation in default only rolls back for unchecked exception such as RuntimeException. So all changes will be reflected to database.

Option A and E is right. More info can be found in https://docs.spring.io/spring-framework/docs/current/javadoc-api/org/springframework/transaction/annotation/Transactional.html

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [ ] It will try to delete the student then the DataIntegrityViolationException will be thrown.

- [ ] It will delete the student successfully then an SQLException will be thrown with the message "Cannot delete all Lessons. There are lessons in progress state".

- [x] ** If @Transactional annotation is put in place of X (StudentServer.java), after saving the student and lessons, it will update lessons and remove the student. Then it will throw an SQLException and all database changes will be rolled back. **

- [ ] It will throw a RuntimeException with the message "Student not found".

- [ ] If @Transactional annotation is put in place of X (StudentServer.java), after saving the student and lessons, it will update lessons and remove the student. Then it will throw an SQLException.

No comments.

3. Configuration Properties

Correct

Multiple Choice | Spring | Spring Boot | Application Properties Annotations | Medium

Question description

An *application.properties* file contains the following properties:

```
app.prop1 = value1
app.prop2 = value2
```

You decided to create a class to hold this configuration. To protect it, the class is defined as final.

```
@Configuration
@ConfigurationProperties(prefix = "app")
public final class AppConfiguration {
  String prop1;
  String prop2;

  public String getProp1() {
    return prop1;
  }

  public void setProp1(String prop1) {
    this.prop1 = prop1;
  }
```

```
  public String getProp2() {
    return prop2;
  }

  public void setProp2(String prop2) {
    this.prop2 = prop2;
  }
}
```

What is the result of running the Spring application with this class?

Interviewer guidelines

Spring creates dynamic proxies for classes annotated with `@Configuration` classes. Spring uses CGLIB to extend your class to create proxy. Hence, configuration classes cannot be final. When running the code Spring throws `BeanDefinitionParsingException` and this is the correct option.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ The application runs, the configuration bean is created. and the properties are initialized correctly (prop1=value1, prop2=value2).
- ○ The application runs, and a configuration bean is created, but properties are set to null (prop1=null, prop2=null).
- ● ** A BeanDefinitionParsingException is thrown because the @Configuration bean cannot be marked as final. ** ✓

An IllegalStateException is thrown because the @Configuration bean cannot have an @ConfigurationProperties annotation.

A NoSuchMethodException is thrown because the @Configuration bean needs a default constructor.

No comments.

4. Endpoint Security

Correct

Multiple Choice | Spring Boot | Medium

Question description

In Spring Boot Security, the option to enable secure endpoints with @PreAuthorize annotation is:

Interviewer guidelines

@EnableGlobalMethodSecurity by default doesn't enable this feature. To enable this prePostEnabled=true value must be set

Candidate's Solution

Options: (Expected answer indicated with a tick)

Configuration class annotated with @EnableGlobalMethodSecurity

Configuration class annotated with @EnableGlobalMethodSecurity(securedEnabled = true)

- [x] Configuration class annotated with @EnableGlobalMethodSecurity(prePostEnabled = true)
- [ ] None above

No comments.

5. Predict the Output

Correct

Multiple Choice | Medium | Exception Handling | Spring Boot | Aspect-Oriented Programming

Question description

Consider the following Snippet and predict the output.

The following API is triggered: http://localhost:8080/add/employee?eId=123&fName=Ravi&lastName=Shekhar

```
@Autowired
private EService eService;

@RequestMapping(value = "/add/employee", method = RequestMethod.GET)
public com.model.Employee add(@RequestParam("eId") String eId, @RequestParam("fName") Strin
g fName, @RequestParam("lastName") String lastName)
{
   return eService.createEmployee(eId, fName, lastName);
}

@Aspect
@Component
public class EServiceAspect
{
   @After(value = "execution(* com.random.service.EService.*(..)) && args(eId, fName, lastName)")
   public void afterAdvice(JoinPoint joinPoint, String eId, String fName, String lastName) {
```

```
        System.out.println("After method:" + joinPoint.getSignature());
        System.out.println("Creating Employee with first name - " + fName+ ", second name - " + lastNa
me+ " and id - " + eId);
    }
}
```

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [x] ** After method : com.test.model.Employee addEmployee(String,String,String) Creating Employee with first name - Ravi , second name - Shekhar and id - 123   **

- [ ] After method : com.test.model.Employee addEmployee(string,string,string) Creating Employee with first name - Ravi , second name - Shekhar and id - 123

- [ ] After method : com.test.model.Employee addEmployee(empId,firstName,secondName) Creating Employee with first name - Ravi , second name - Shekhar and id - 123

- [ ] After method : com.test.model.Employee addEmployee(String,String,String) Creating Employee with first name - Shekhar, second name - Ravi and id - 123

No comments.

6. Wind Farms

Correct

Coding | Implementation | Medium | Arrays | Algorithms | Problem Solving

Question description

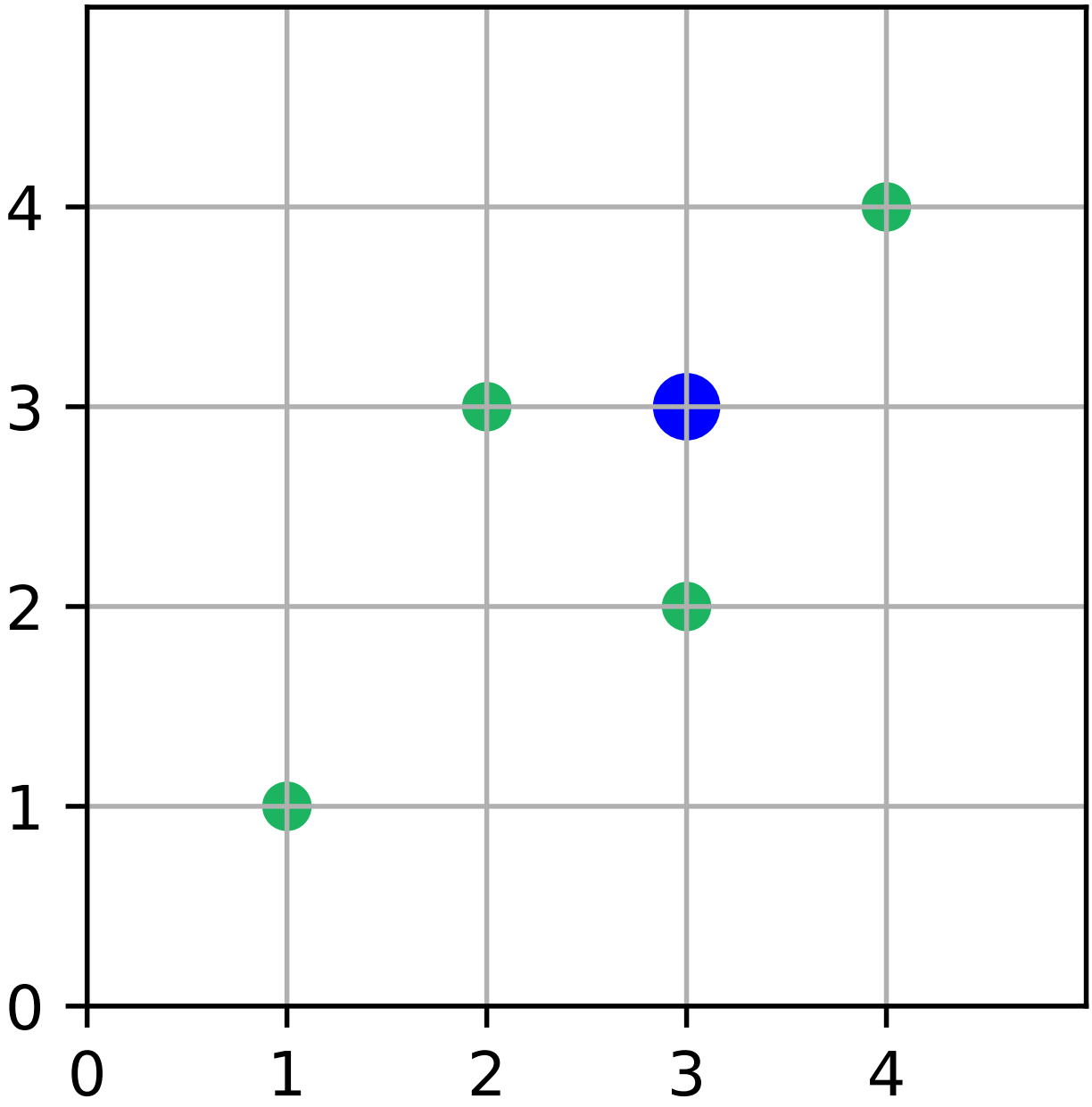
An energy company needs to place a control center to manage a distributed wind farm. Your task is to determine the optimal location for the control center that minimizes the total connection cost.

For each wind farm *i*:

- The farm is located at coordinates *(x[i], y[i])*
- The control center is at coordinates *(cx, cy)*
- The distance is calculated as *|x[i] - cx| + |y[i] - cy|*
- The connection cost is *premium[i] * distance*

Example

x = [1, 3, 2, 4]
y = [1, 2, 3, 4]
premium = [1, 3, 2, 4]



The optimal location is at position [3, 3], resulting in:

Farm #	Location	Distance	Premium	Cost
0	(1, 1)	|1 - 3| + |1 - 3| = 4	1	4 * 1 = 4

1	(3, 2)	|3 - 3| + |2 - 3| = 1	3	1 * 3 = 3
2	(2, 3)	|2 - 3| + |3 - 3| = 1	2	1 * 2 = 2
3	(4, 4)	|4 - 3| + |4 - 3| = 2	4	2 * 4 = 8

The total connection cost is 17, which is the minimum possible.

Function Description

Complete the function *windFarms* in the editor with the following parameter(s):

int premium[n]: the distance multiplier for the cost to connect
int x[n]: *x* coordinates of the wind farms
int y[n]: *y* coordinates of the wind farms

Returns

int: the minimum cost to connect the wind farms

Constraints

- $1 \le n \le 10^3$
- $1 \le x[i], y[i] \le 10^4$
- $1 \le premium[i] \le 50$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains an integer *n*, the size of the array *premium*.
The next *n* lines each contain an element, *premium[i],* where 0 ≤ i < n.
The next line contains an integer *n*, the size of *x[]*.
The next *n* lines each contain an element, *x[i]*.
The next line contains an integer *n*, the size of *y[]*.
The next *n* lines each contain an element, *y[i]*.

Note: *n* is the same value in each instance.

▼ SAMPLE CASE 0

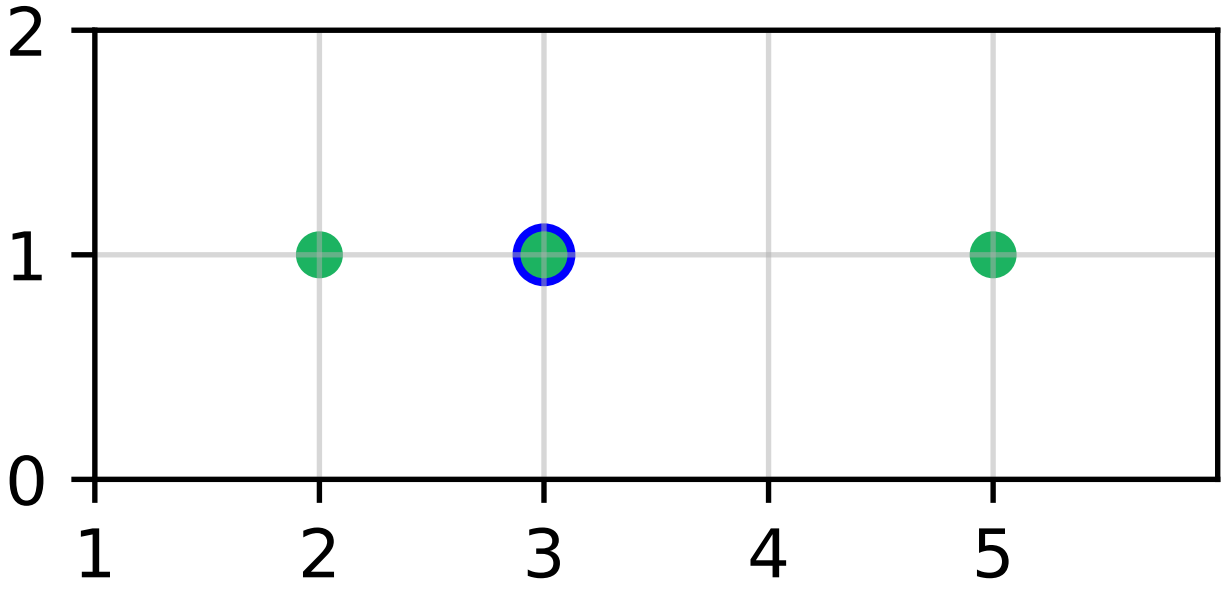
Sample Input 0

```
STDIN    Function
-----    --------
3    →  premium[] size n = 3
1    →  premium = [1, 1, 1]
1
1
3    →  x[] size n = 3
2    →  x = [2, 3, 5]
3
5
3    →  y[] size n = 3
1    →  y = [1, 1, 1]
1
1
```

Sample Output 0

```
3
```

Explanation 0



Locate the controller at farm 1.

Farm #	Location	Distance	Premium	Cost
0	(1, 1)	|1 - 3| + |1 - 1| = 1	1	1 * 1 = 1
1	(3, 1)	|3 - 3| + |1 - 1| = 0	1	0 * 1 = 0

2	(5, 1)	|5 - 3| + |1 - 1| = 2	1	2 * 1 = 2

▼ SAMPLE CASE 1

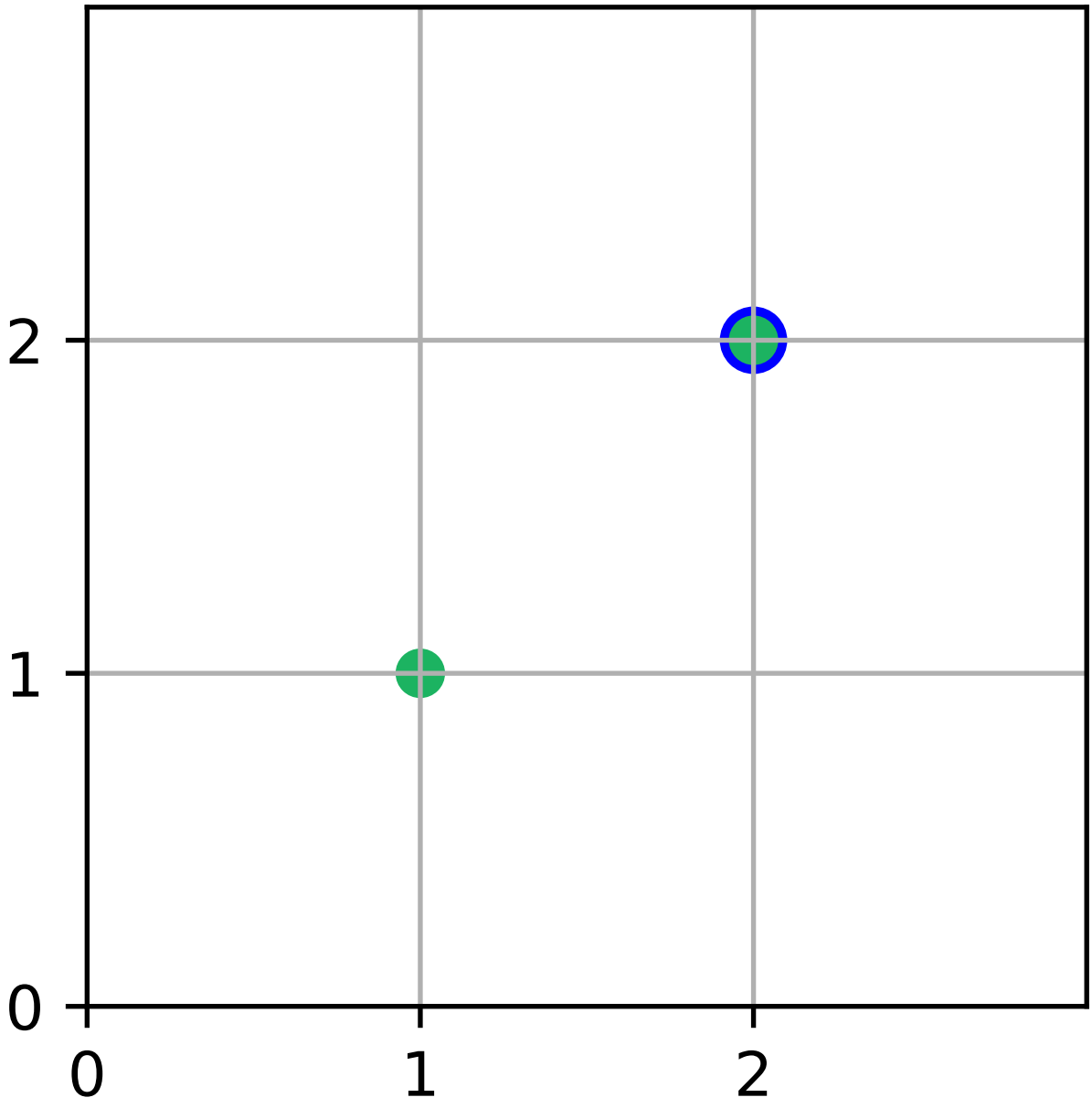
Sample Input 1

```
STDIN    Function
-----    --------
2    →  premium[] size n = 2
1    →  premium = [1, 10]
10
2    →  x[] size n = 2
1    →  x = [1, 2]
2
2    →  y[] size n = 2
1    →  y = [1, 2]
2
```

Sample Output 1

```
2
```

Explanation 1



The sum of the distances to a center at (1, 1), (1, 2), (2, 2), or (2, 1) is 2. Since the premium to connect farm 1 is 10, it is best to locate at position (2, 2) for a cost of 2 * 1 + 0 * 10 = 2.

▼ SAMPLE CASE 2

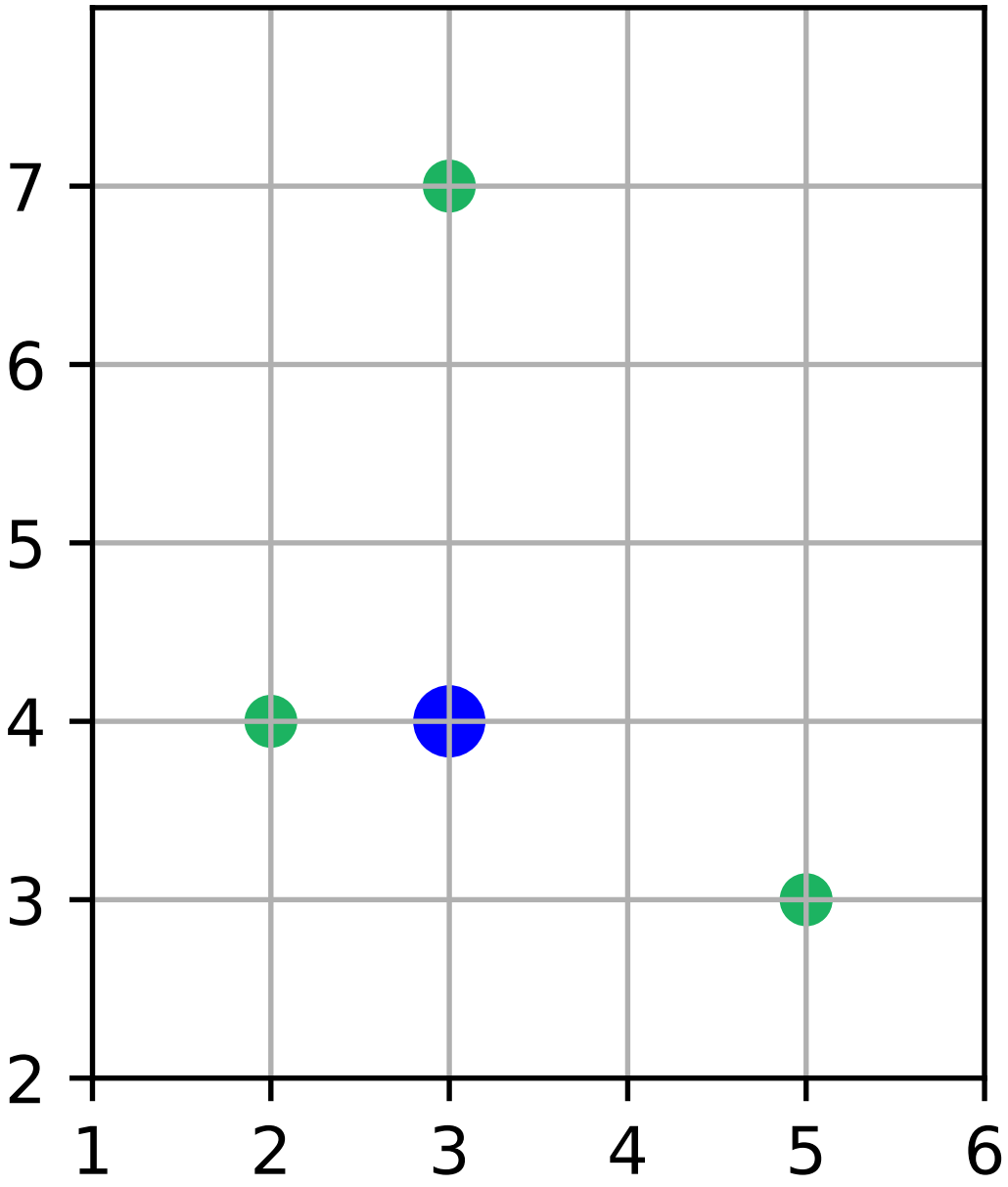
Sample Input 2

```
STDIN    Function
-----    --------
3    →  premium[] size n = 3
1    →  premium = [1, 1, 1]
1
1
3    →  x[] size n = 3
5    →  x = [5, 2, 3]
2
3
3    →  y[] size n = 3
3    →  y = [3, 4, 7]
4
7
```

Sample Output 2

```
7
```

Explanation 2



The optimal location is (3, 4) for a cost of 3 + 1 + 3 = 7.

Interviewer guidelines

▼ SOLUTION

Skills: Loops and counters, I/O

The cost can be calculated for x and y coordinates separately. The function examines each possible x and y coordinate for the control center separately (from 1 to 10000), computes the total cost of connecting all farms for each possible coordinate, and captures the minimum cost found. The ultimate minimum cost is the summation of these minimum costs for x and y coordinates separately.

Optimal Solution

```
def windFarms(premium, x, y):
    minCostX = 10**10
    n = len(x)

    for cx in range(1, 10001):
        cost = 0
        for i in range(n):
            cost += premium[i] * abs(cx - x[i])

        minCostX = min(minCostX, cost)

    minCostY = 10**10
    for cy in range(1, 10001):
        cost = 0
        for i in range(n):
            cost += premium[i] * abs(cy - y[i])

        minCostY = min(minCostY, cost)

    return minCostX + minCostY
```

▼ **COMPLEXITY ANALYSIS**

Time Complexity - O(n)

The universe of x and y values are tested against 10000 values.

Space Complexity - O(1)

Constant space is required regardless of n.

Candidate's Solution

Language used: **Java 8**

```
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


class Result {

    /*
     * Complete the 'windFarms' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts following parameters:
     *  1. INTEGER_ARRAY premium
     *  2. INTEGER_ARRAY x
     *  3. INTEGER_ARRAY y
     */

    public static long windFarms(List<Integer> premium, List<Integer> x, List<Integer> y) {
    // Write your code here
    int cx= weightedMedian(x, premium);
    int cy= weightedMedian(y, premium);

```

```
    long totalCost=0;
    for(int i=0; i<premium.size(); i++){
       // totalCost+=premium.get(i)*(Math.abs(x.get(i)-cx))+(Math.abs(y.get(i)-cy));

       long distance=Math.abs(x.get(i)-cx)+(Math.abs(y.get(i)-cy));
       totalCost += (long) premium.get(i)*distance;
    }
    return totalCost;
     }
    private static int weightedMedian (List<Integer>coords, List<Integer>weight){
        int n=coords.size();
        List<int[]>list= new ArrayList<>();

        for(int i=0; i<n; i++){
            list.add(new int[]{coords.get(i),weight.get(i)});
        }
        list.sort(Comparator.comparingInt(a->a[0]));
        int totalWeight=0;
        for(int w: weight)totalWeight+=w;
        int target=(totalWeight+1)/2;

        int cumulative=0;
        for(int[] pair : list){
            cumulative+=pair[1];
            if(cumulative>= target){
                return pair[0];
            }
        }
        return 0;

    }

}
public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int premiumCount = Integer.parseInt(bufferedReader.readLine().trim());

```

```
        List<Integer> premium = IntStream.range(0, premiumCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int xCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> x = IntStream.range(0, xCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int yCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> y = IntStream.range(0, yCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        long result = Result.windFarms(premium, x, y);

        bufferedWriter.write(String.valueOf(result));
       // bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
```

```
}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int premiumCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> premium = IntStream.range(0, premiumCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int xCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> x = IntStream.range(0, xCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int yCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> y = IntStream.range(0, yCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
```

```
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int result = Result.windFarms(premium, x, y);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
TestCase 0	Easy	Sample	Success	1	0.1155 sec	30.4 KB
TestCase 1	Easy	Sample	Success	1	0.1035 sec	30.5 KB
TestCase 2	Easy	Sample	Success	1	0.153 sec	30.5 KB
TestCase 3	Easy	Sample	Success	2	0.1126 sec	30.8 KB
TestCase 4	Easy	Hidden	Success	2	0.1066 sec	30.8 KB
TestCase 5	Easy	Hidden	Success	2	0.1129 sec	30.6 KB
TestCase 6	Medium	Sample	Success	6	0.1033 sec	31.1 KB
TestCase 7	Medium	Hidden	Success	6	0.1215 sec	30.8 KB

TestCase 8	Medium	Hidden	Success	6	0.1075 sec	31 KB
TestCase 9	Hard	Hidden	Success	12	0.153 sec	34.9 KB
TestCase 10	Hard	Hidden	Success	12	0.1441 sec	33.3 KB
TestCase 11	Hard	Hidden	Success	12	0.2263 sec	34 KB
TestCase 12	Hard	Hidden	Success	12	0.1545 sec	33.8 KB

No comments.

7. Java: DataVille's Optimal Storage

Correct

Coding Greedy Sorting Medium Java

Question description

In the digital city of DataVille, efficient data management is critical. The city's database is represented as a binary array, *databaseStorage*, containing *n* bits of data. Each bit represents one of two types of data:

- Standard data: Represented by 0.
- Priority data: Represented by 1.

To optimize efficiency during peak operational hours, the number of transitions between 0s and 1s in the array must be minimized. Transitions are points in the array where adjacent elements differ (e.g., a 0 followed by a 1 or a 1 followed by a 0).

Operations:

- You can swap any two bits in the array as many times as you like.

The goal is to rearrange the bits to minimize transitions with the minimum number of swaps.

Example

Consider that *databaseStorage* = [1, 0, 1, 0, 1]. One optimal strategy is to swap *databaseStorage[0]* and *databaseStorage[3]*. After performing this operation, the array is [**0**, 0, 1, **1**, 1], which requires only one transition between *databaseStorage[1]* and *databaseStorage[2]*. This requires only 1 swap, so the return value is 1.

Function Description

Complete the function g*etOptimalStorage* in the editor below.

Function Parameters

int databaseStorage[n]: the current organization of the database's storage, with 0s for standard data and 1s for priority data

Returns

int: the minimum number of operations required to minimize transitions between 0s and 1s

Constraints

- $2 \le n \le 10^5$
- *databaseStorage[i]* $\in$ {0, 1}

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains an integer, *n,* the size of *databaseStorage*.
Each of the next lines contains an integer, *databaseStorage[i].*

▼ SAMPLE CASE 0

Sample Input 0

```
STDIN       FUNCTION
-----       --------
7           databaseStorage[] size n = 7
1           databaseStorage = [1, 0, 1, 1, 0, 0, 1]
0
1
1
0
```

```
0
1
```

Sample Output 0

```
1
```

Explanation

One optimal strategy is to swap *databaseStorage[1]* (which is 0) with *databaseStorage[6]* (which is 1). After performing this operation, the array becomes [1, **1**, 1, 1, 0, 0, **0**]. This solution requires only 1 swap, so the output is 1.

▼ SAMPLE CASE 1

Sample Input 1

```
STDIN     FUNCTION
-----     --------
9         databaseStorage[] size n = 9
1         databaseStorage = [1, 0, 0, 1, 1, 0, 0, 0, 1]
0
0
1
1
0
0
0
1
```

Sample Output 1

```
2
```

Explanation

One optimal strategy of operations is:

1. Swap *databaseStorage*[1] and *databaseStorage*[8]. After performing this operation, the array becomes [1, **1**, 0, 1, 1, 0, 0, 0, **0**].
2. Swap *databaseStorage*[2] and *databaseStorage*[4]. After performing this operation, the array becomes [1, 1, **1**, 1, **0**, 0, 0, 0, 0].

Now there is only one transition. The minimum number of swaps is 2.

Interviewer guidelines

▼ SOLUTION

The solution is to count the number of 0s and 1s in the array. Then, we count the number of 0s and 1s that are out of place. The answer is the minimum of the two counts.

```
class Result {

    /*
     * Complete the 'getOptimalStorage' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts INTEGER_ARRAY databaseStorage as parameter.
     */

    public static int getOptimalStorage(List<Integer> databaseStorage) {
        int zeros = 0, ones = 0;

        for (int brick : databaseStorage) {
            if (brick == 0) {
                zeros++;
            } else {
                ones++;
            }
        }

        int startZero = 0, startOne = 0;

        for (int i = 0; i < ones; ++i) {
            if (databaseStorage.get(i) != 1) {
                startOne++;
            }
        }

        for (int i = 0; i < zeros; ++i) {
            if (databaseStorage.get(i) != 0) {
                startZero++;
            }
        }

        return Math.min(startZero, startOne);
    }
}
```

▼ COMPLEXITY

Time Complexity - O(n)

To traverse over the array, we use a loop that runs n times. The complexity is O(n).

Space Complexity - O(1)

We use only 2 variables, hence the space complexity is O(1).

Candidate's Solution

Language used: **Java 8**

```
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


class Result {

    /*
     * Complete the 'getOptimalStorage' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts INTEGER_ARRAY databaseStorage as parameter.
     */

    public static int getOptimalStorage(List<Integer> databaseStorage) {
        int n= databaseStorage.size();
        int count1=0;
        for(int num: databaseStorage){
            if(num==1 )count1++;
        }
        int count0= n-count1;
        int swaps1= 0;
        for(int i=0;i<count1; i++){
            if(databaseStorage.get(i)==0){
```

```
                swaps1++;

            }
        }

        int swaps2= 0;
        for(int i=0;i<count0; i++){
            if(databaseStorage.get(i)==1){
                swaps2++;

            }
        }
        return Math.min(swaps1, swaps2);

    }

}
public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int databaseStorageCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> databaseStorage = IntStream.range(0, databaseStorageCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int result = Result.getOptimalStorage(databaseStorage);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
```

```
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
TestCase 0	Easy	Sample	Success	1	0.1157 sec	30.4 KB
TestCase 1	Easy	Sample	Success	1	0.1011 sec	30.5 KB
TestCase 2	Easy	Sample	Success	1	0.1101 sec	30.3 KB
TestCase 3	Easy	Hidden	Success	3	0.1357 sec	30.4 KB
TestCase 4	Easy	Hidden	Success	3	0.0997 sec	30.6 KB
TestCase 5	Easy	Hidden	Success	3	0.1058 sec	29.8 KB
TestCase 6	Easy	Hidden	Success	4	0.1045 sec	30.3 KB
TestCase 7	Easy	Hidden	Success	4	0.0966 sec	30.3 KB
TestCase 8	Easy	Hidden	Success	5	0.1273 sec	30.4 KB
TestCase 9	Easy	Hidden	Success	5	0.1169 sec	30.3 KB
TestCase 10	Easy	Hidden	Success	5	0.1057 sec	29.8 KB

TestCase 11	Medium	Hidden	Success	7	0.1263 sec	31.3 KB
TestCase 12	Medium	Hidden	Success	8	0.1311 sec	31.3 KB
TestCase 13	Hard	Hidden	Success	10	0.1162 sec	31.3 KB
TestCase 14	Hard	Hidden	Success	15	0.3453 sec	55.3 KB

No comments.